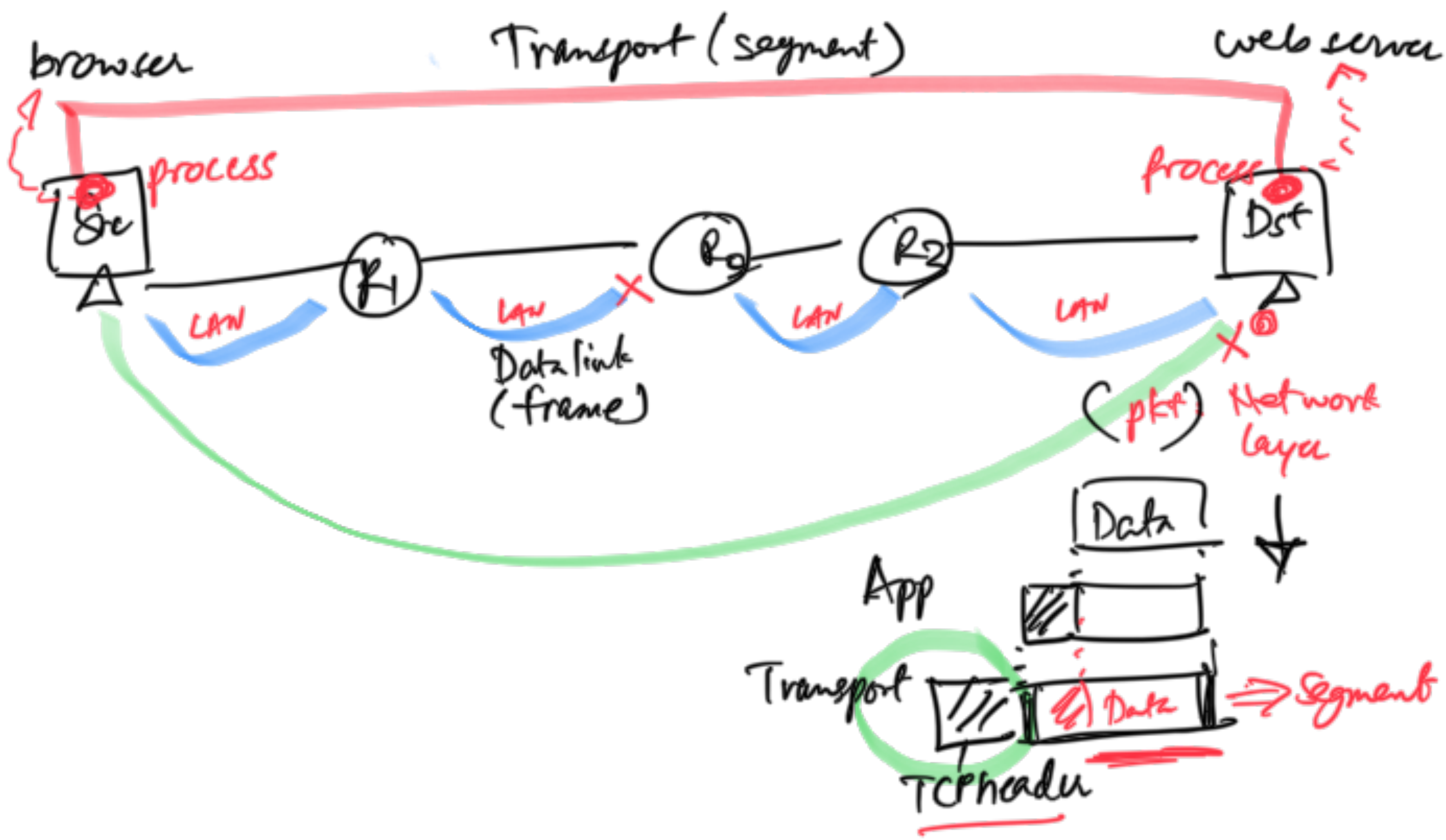
Lec 14 : Transmission Control Protocol (TCP)

9/10/23

Recap.

- Reliable data transfer approaches
  1. Stop & wait – send one pkt, stop & wait for ACK, send next...
  2. Sliding window OR pipelining – can send window size no. of packets before waiting for an ACK
     - Error recovery approaches
       a. Go-back-N (GBN)
       b. Selective repeat (SR)



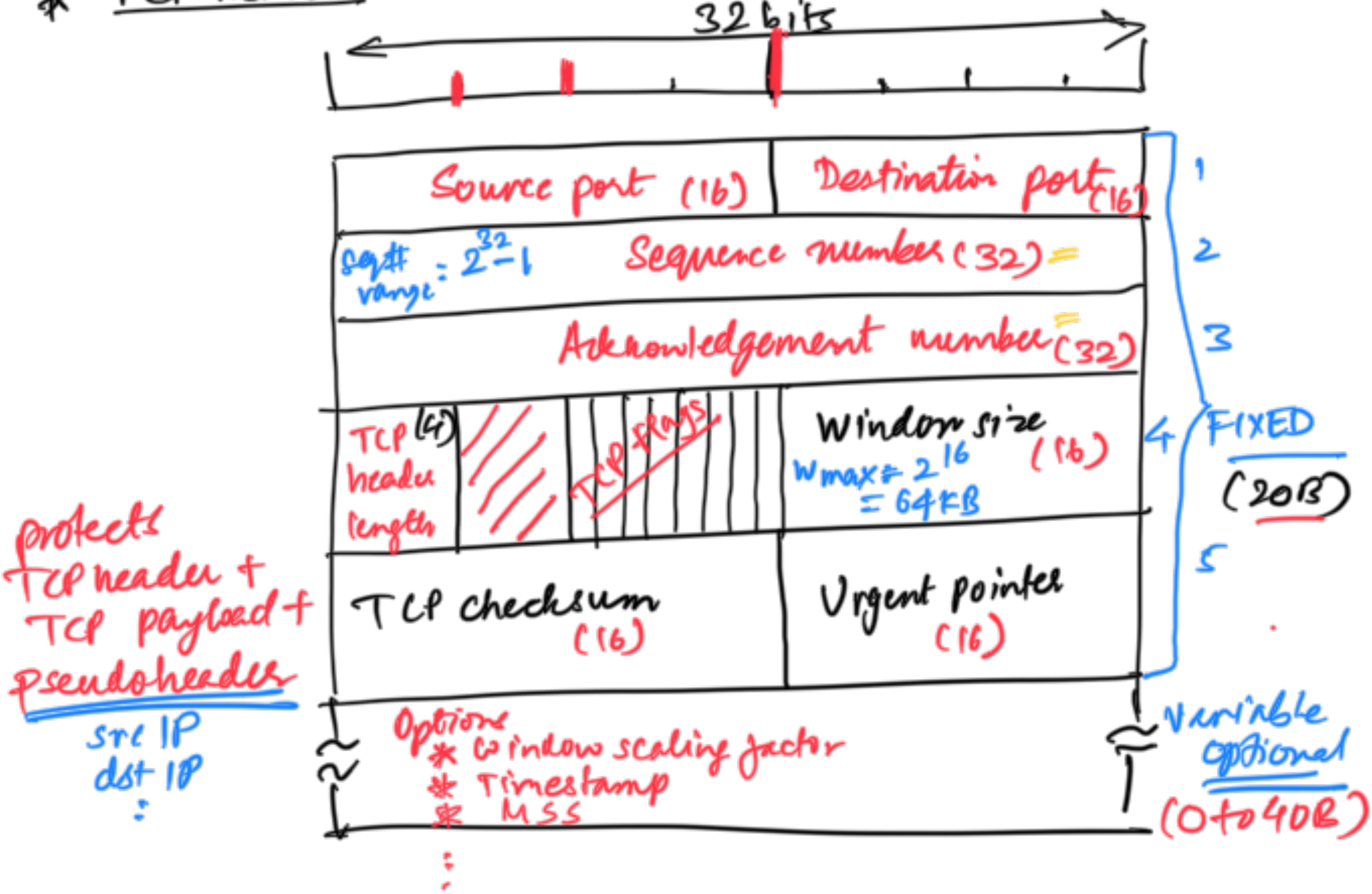
* TCP (Transmission Control Protocol)

Functions

- process-to-process delivery ✓
- Connection-oriented service ✓
  - → Initial handshake (connⁿ estab)
  - → Data transfer
  - → Terminate connⁿ
- Reliable delivery
  - → GBN
  - → SR
- Flow control — matching RX speed with the TX (buffer read speed); RX tells the TX speed at which TX should send
- Congestion control
- segmentation & reassembly

* full duplex commⁿ

* TCP header



TCP header length: (4 bits)

length of TCP header in words (1 word = 32 bits = 4B)

THL_{min} = ~~0000~~ 0101 = 5 words = 5 * 4B = 20B

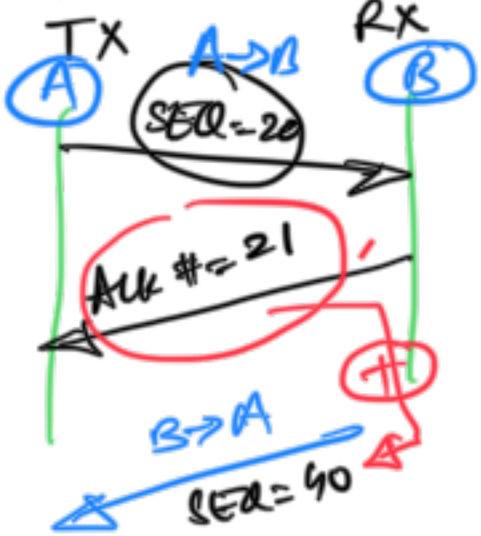
THL_{max} = 1111 = 15 words = 15 * 4B = 60B

TCP flags: (8 bits)

CWR
ECE

congestion control
Explicit congestion notification

RX TX



appending Ack to data pkt
=> piggybacking

URG = Urgent flag
URG=1 => Urgent ptr is valid
Urgent ptr: offset in the segment where urg data ends

Ack
Ack=1 => Ack no. is valid => piggybacking is used
Ack=0 => Ack # can be ignored

PSH
PSH=1 => Rx should not buffer; immediately deliver
PSH=0 => Rx may buffer => efficiency

RST
Reset
RST=1 => reset the connⁿ.
- some problem due to which TCP connⁿ is terminated
- client/server crashes ...

SYN = 1 => Connⁿ initiation req

FIN = 1 => connⁿ termination req
gracefully terminate the connⁿ

TX

A

RX

B

file upload

M1: 20B

M2: 3B

M3: 100B

M4: 40B

M1: 100B

M2: 500B

⋮

| M1 | M2 | ... |

buffer size

| Eth | IP | TCP | ~~M1~~ | Eth |

3B

sender/receiver

* do not buffer

* PSH = 1
push the data as soon as generated

* immediate data delivery

sender buffers data

* PSH = 0

* receiver may buffer
. not deliver it to process immediately